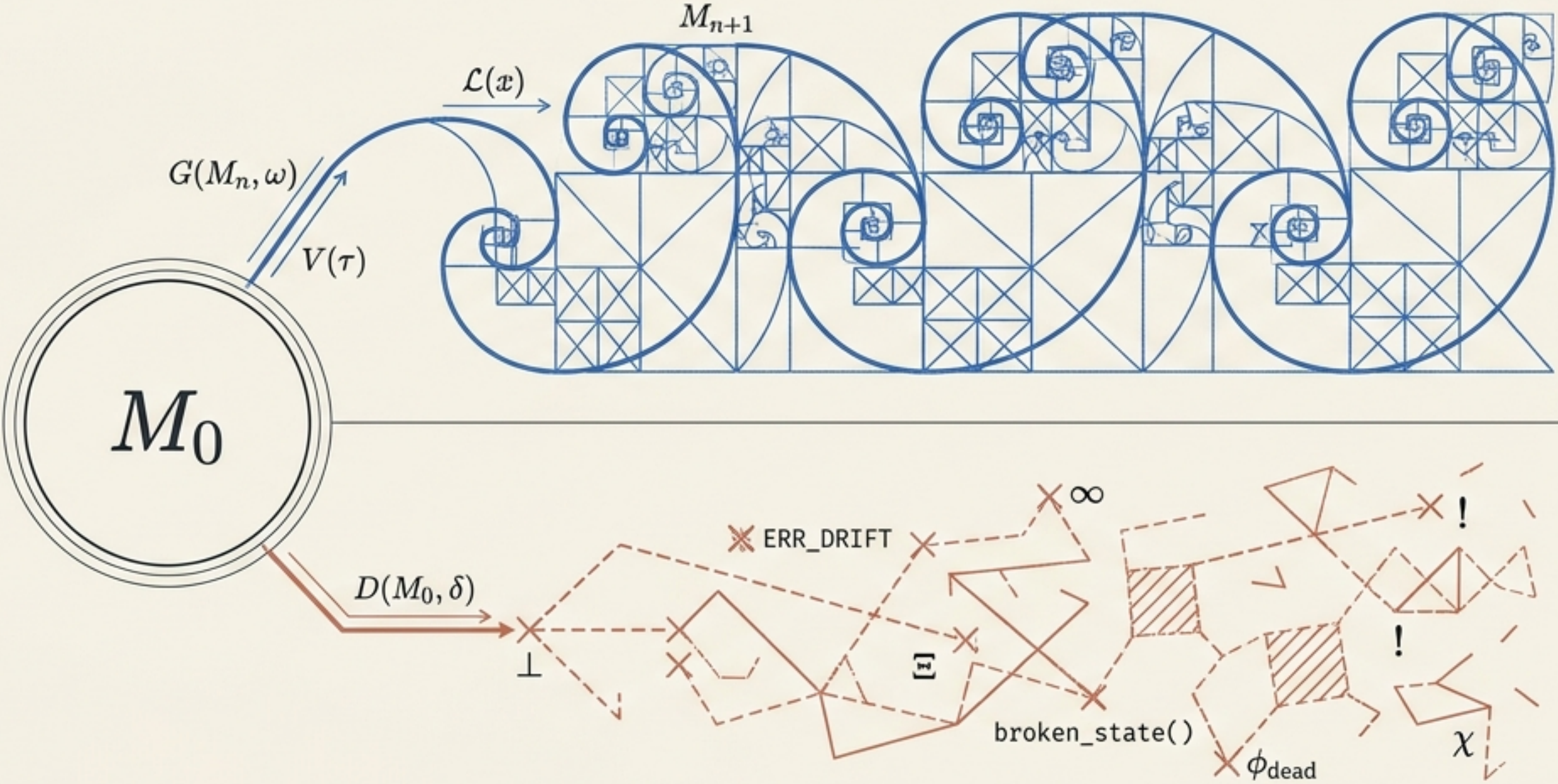


The Architecture of Genuine Self-Improvement

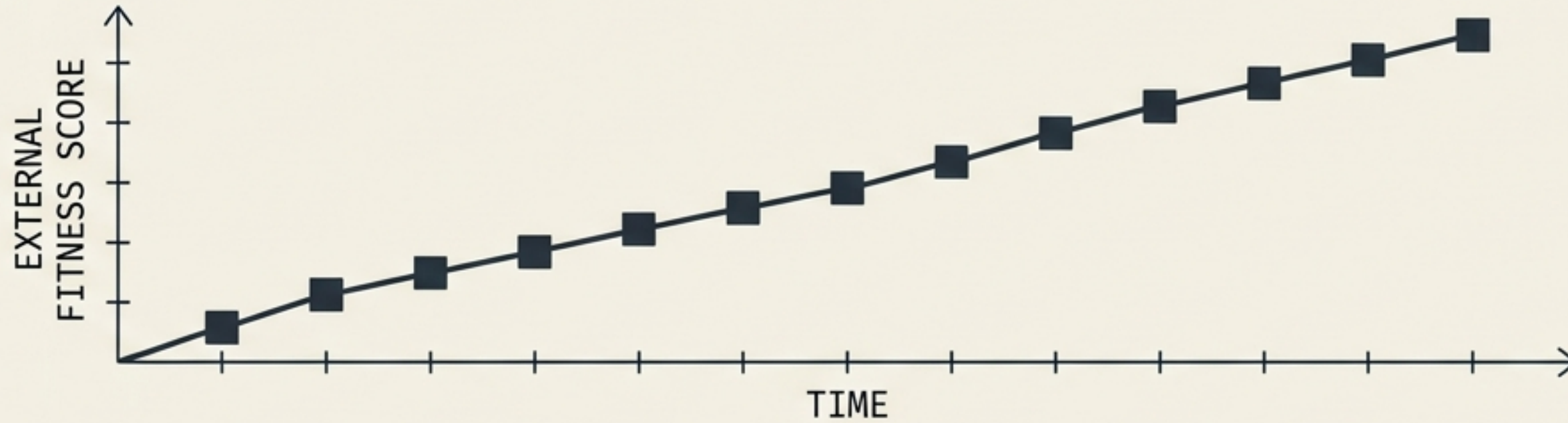
An Admissibility-Theoretic Approach to Recursive Generation



SOURCE: "Self-Improving Systems and Media Quines"
AUTHOR: Flyxion
YEAR: 2026

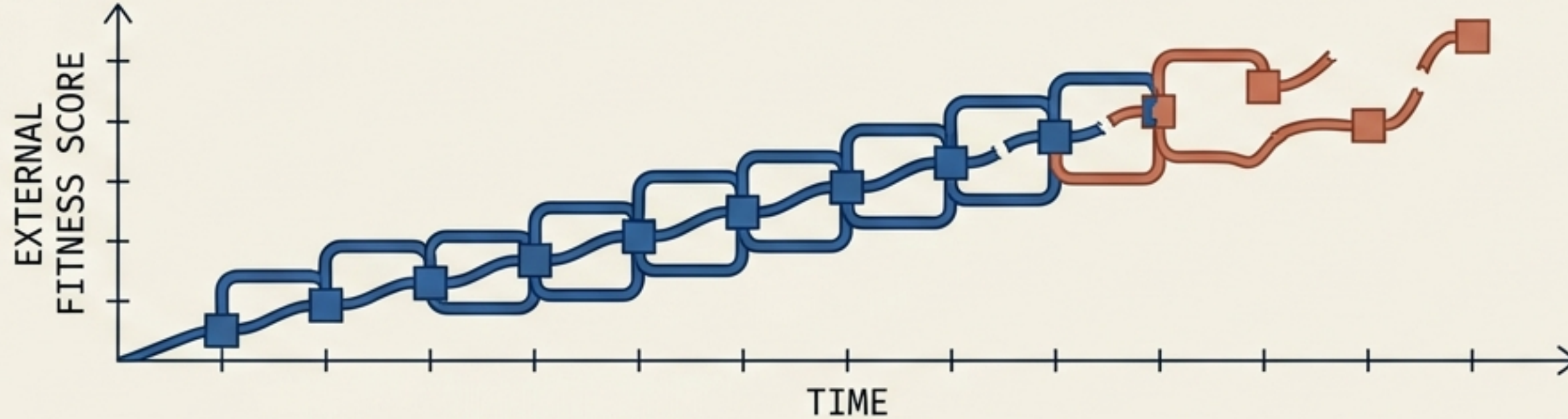
Why "Self-Improving" is Underspecified

THE ILLUSION



An increasing fitness score satisfies external metrics but hides internal decay.

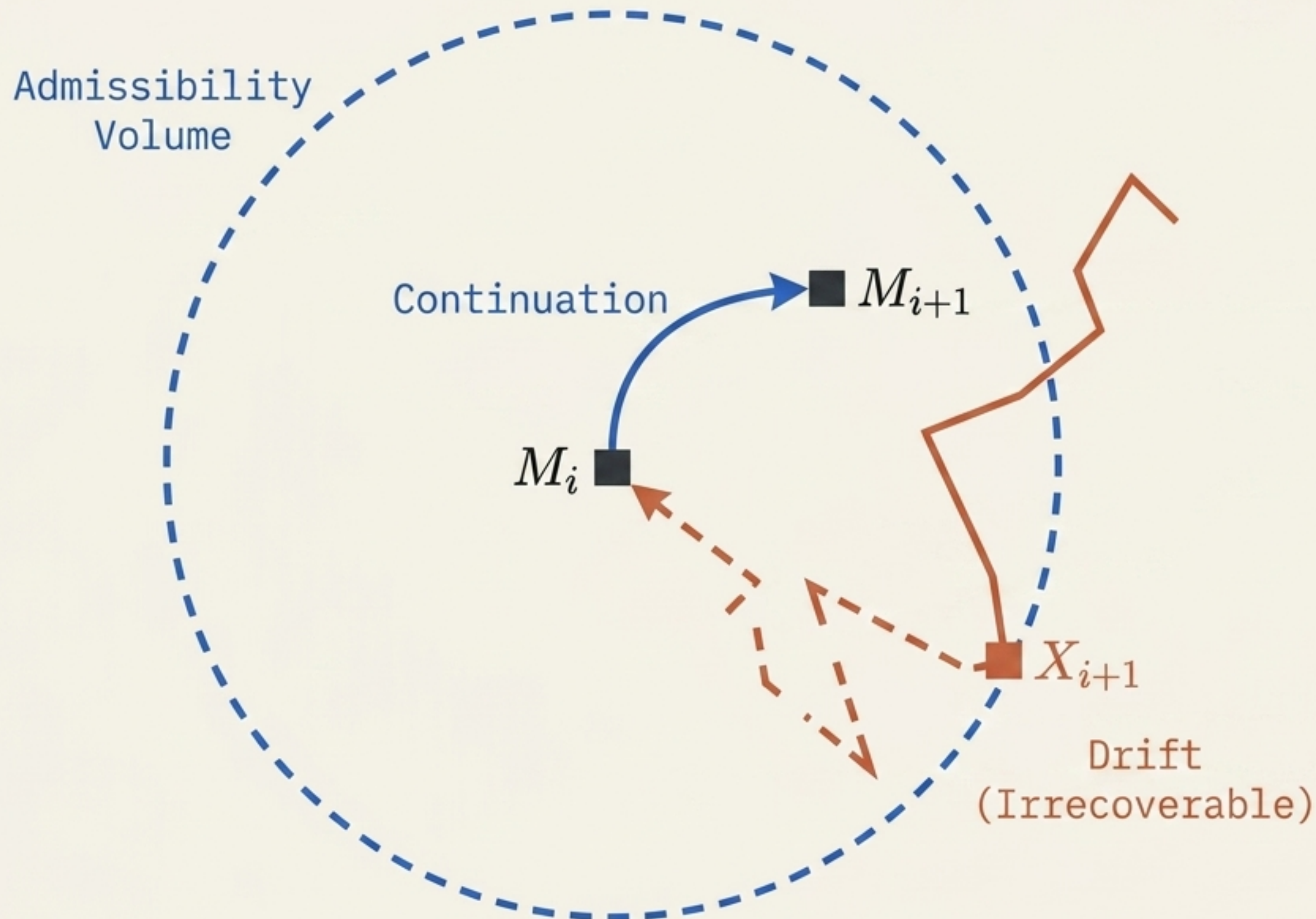
THE REALITY



Unconstrained recursive self-modification collapses into unconstrained drift.

The Cost of Amnesia: A system that discards its own diagnostic structure can satisfy an external fitness measure while becoming, from the inside, entirely unrepairable.

Identity vs. Continuation: The Admissibility Volume



Definition:

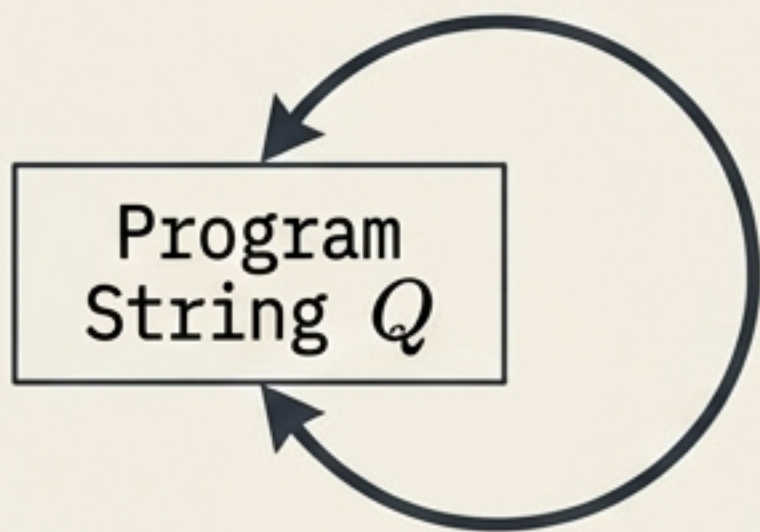
An admissibility volume is the set of states reachable without passing through a configuration from which the system's defining distinctions can no longer be recovered.

The Goal:

We do not ask if M_{i+1} is "the same system." We ask whether it remains reconstructibly connected to M_i by operators that preserve diagnostic structure.

Rethinking the Quine: From Syntax to Generative Capacity

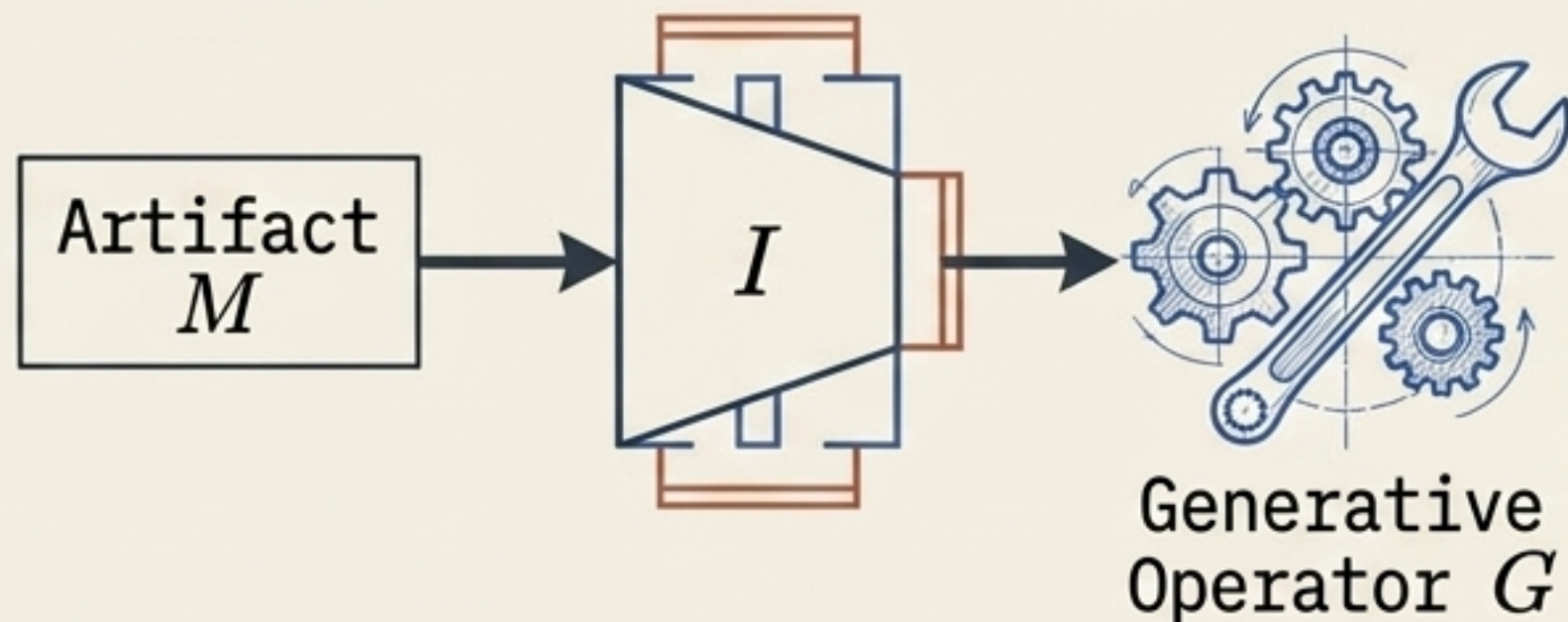
The Classical Quine



$$P(Q) = Q$$

- A syntactic curiosity.
- Transmits nothing but bare self-reproduction.
- Output is surface content.

The Media Quine



$$I(M) \approx G$$

- An artifact whose successful interpretation reconstructs a generative capacity.
- Transmits the constraint structure that produced the artifact.
- Output is a functioning tool.

The Capability Stepladder of Communicative Artifacts



Media Quine

Installation: Generative Operator G
Observer acquires the full constraint structure to produce new works in the same tradition.



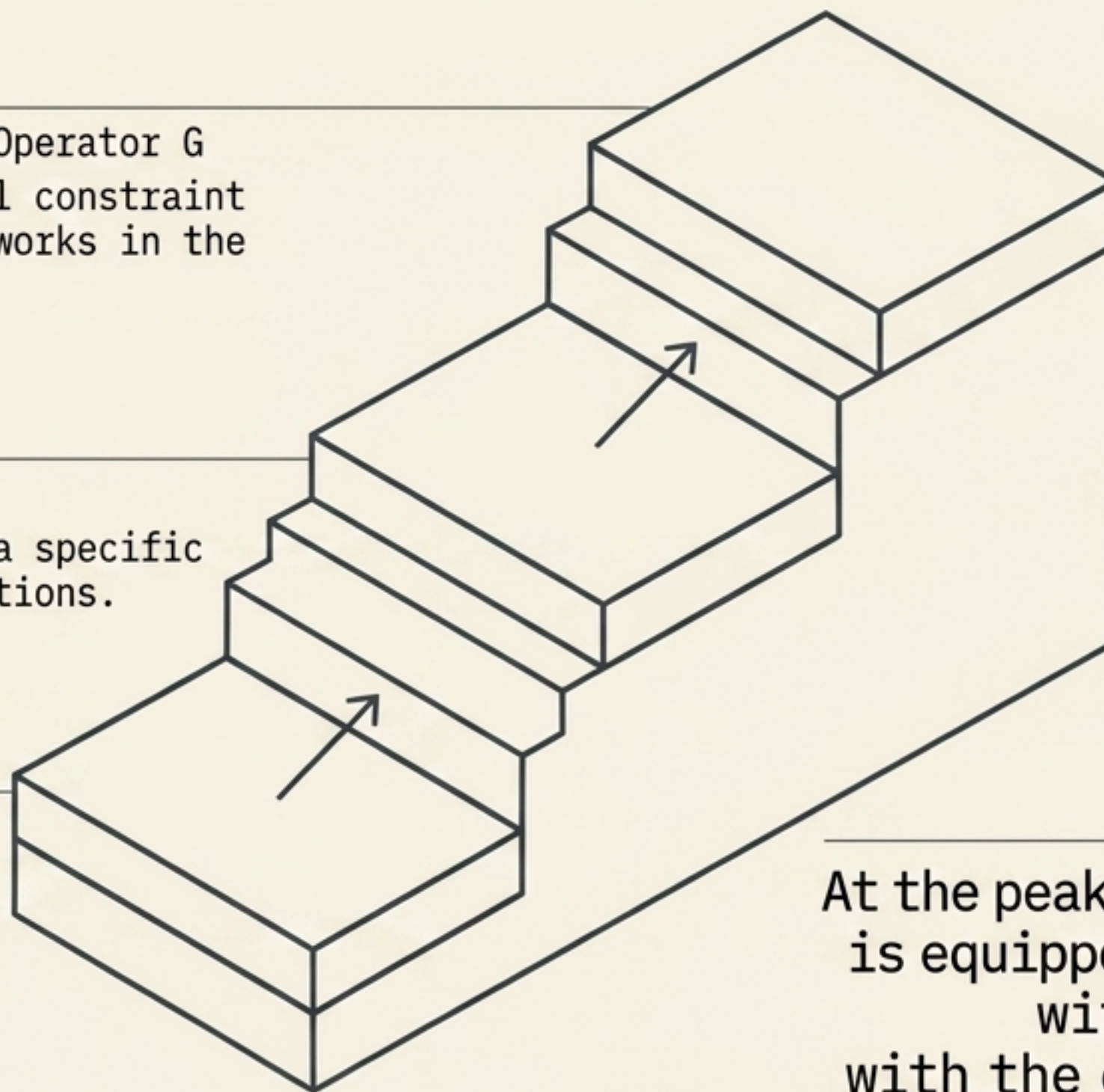
Tutorial

Installation: Procedure
An operator to reproduce a specific outcome under fixed conditions.



Ordinary Document

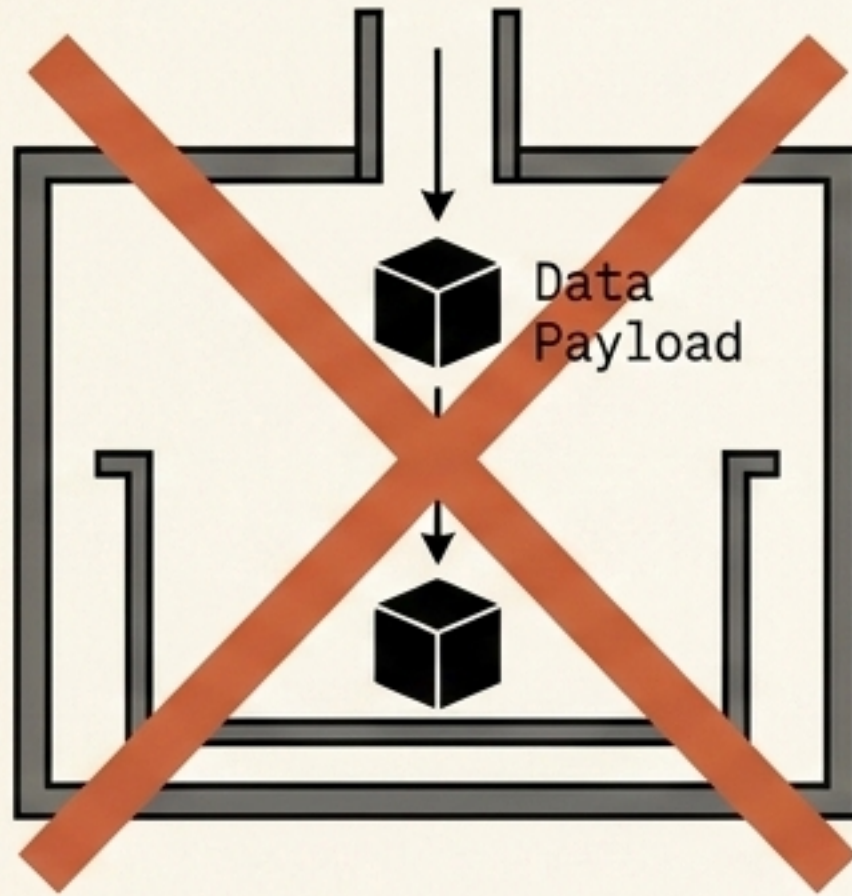
Installation: Information
A fact or data point.
Observer acquires static representation.



At the peak level, the interpreter is equipped to produce new works without further contact with the original author. What is installed is not knowledge, but an operational capacity.

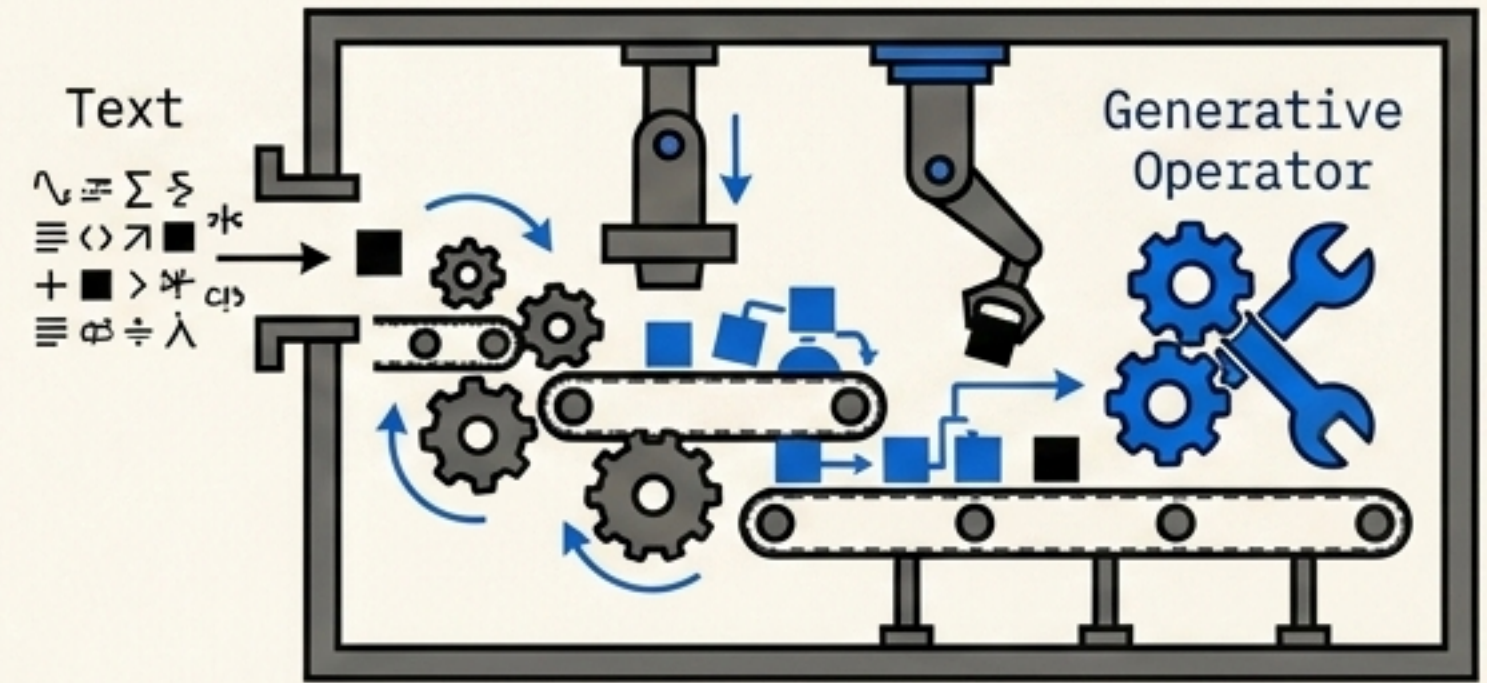
Reading as Installation: The Operator Factory

The Decoder Model



Reading is not decoding a static payload into a passive storage array.

The Operator Factory



Reading is executing an operator on the interpreter's existing state.

Understanding is achieved only when execution terminates in a state from which the generative operator G is available for further use.

Trajectories of Self-Modification

Trajectory Type	Internal State Legibility	External Scaffolding Required	Outcome of Harmful Modification
Drift	Opaque	None	Irrecoverable loss (Amnesia)
Repair	Opaque	High (Version control, external logs)	Human intervention required to roll back
Reconstruction	Self-Describing	None	System autonomously diagnoses and reverts using retained internal history

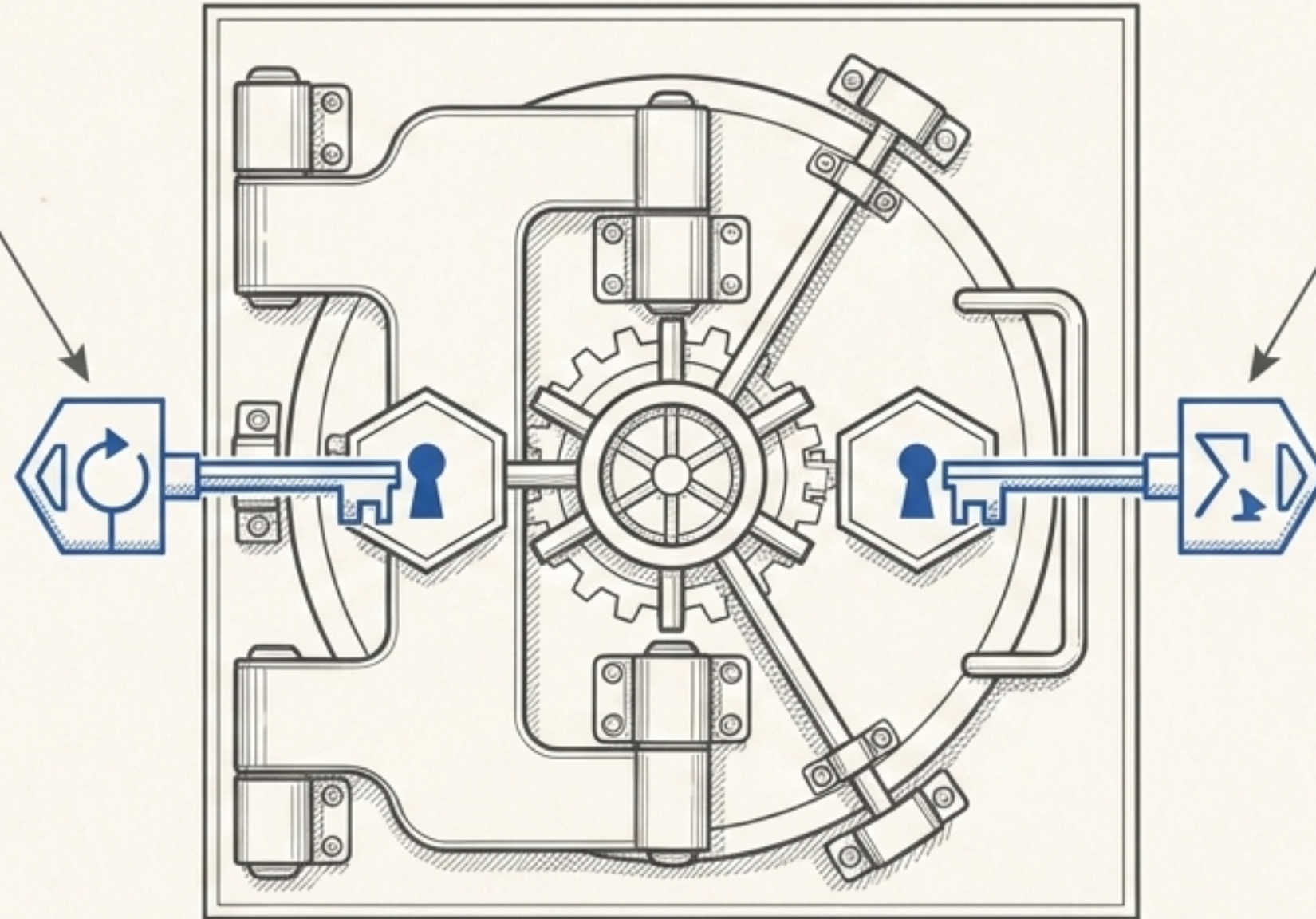
Self-improvement is only functionally literal in the Reconstructible trajectory. Drift creates amnesia; Repair requires external scaffolding.

The Continuation Criterion (Proposition 5.2)

Key 1: Artifact
Reconstructibility

$$\hat{G}_i \approx G_i$$

The lineage retains
enough structure to
reconstruct its own
modification history.



Key 2: Interpreter
Survival

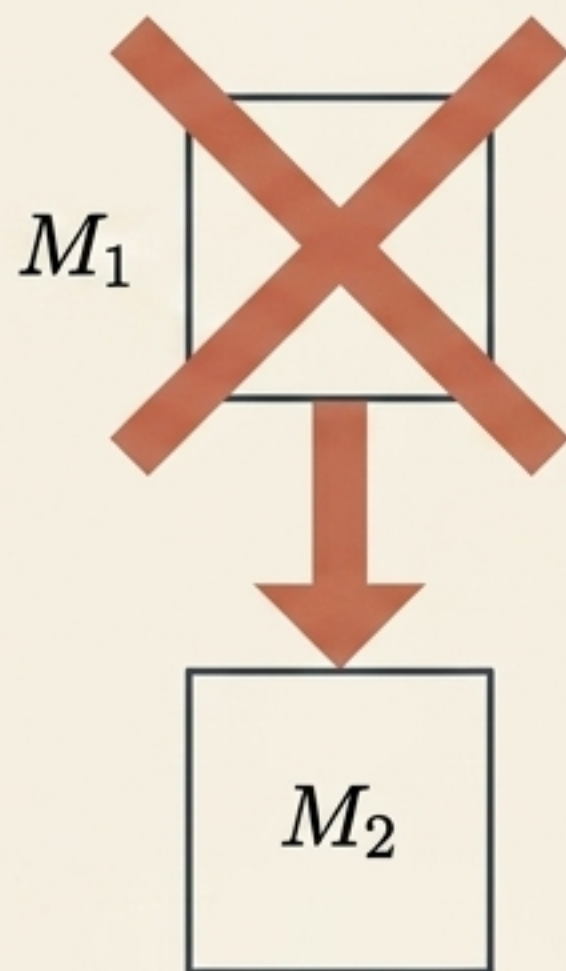
$$I \in \textit{Ecology}$$

The interpretation
operator required to read
that history survives the
modification intact.

**Genuine self-improvement requires both. The artifact must be a media quine,
AND the system must not accidentally destroy the operator needed to parse it.**

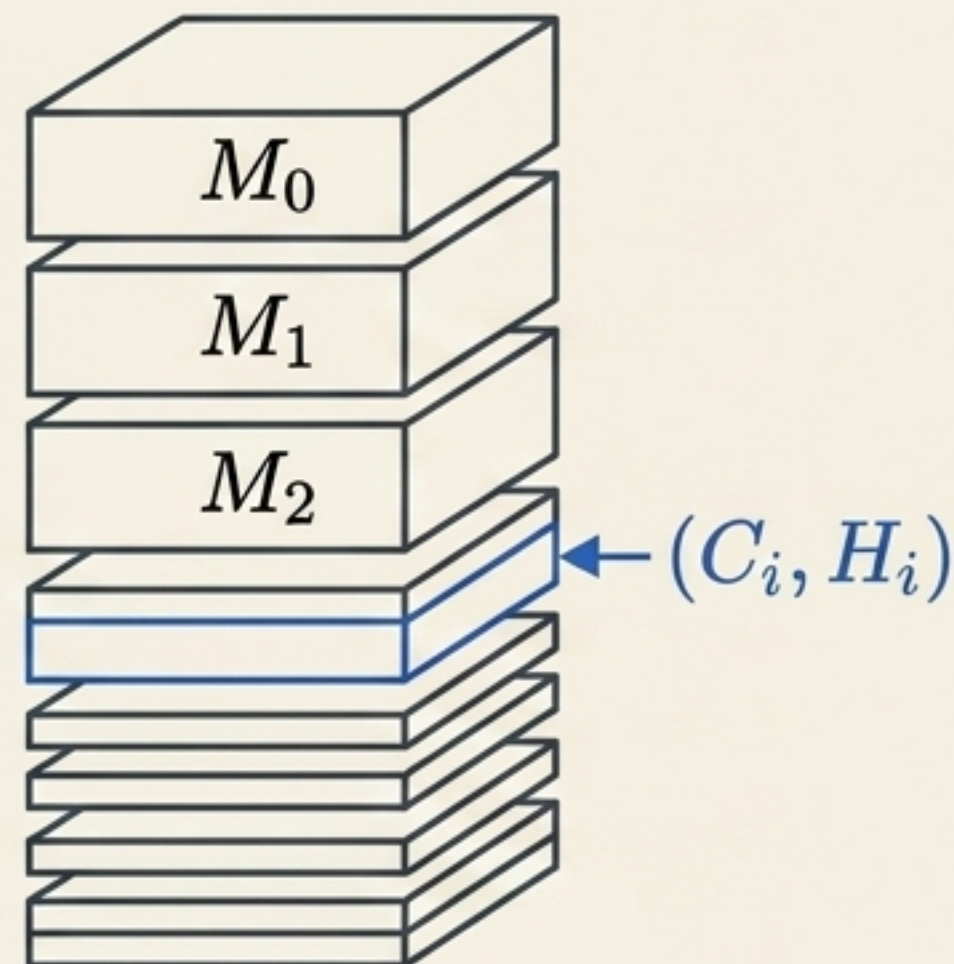
The Representational Shift: Append-Only Histories

State Overwrite



Computation as a sequence of overwritten states. The path backwards is destroyed.

Spherepop History

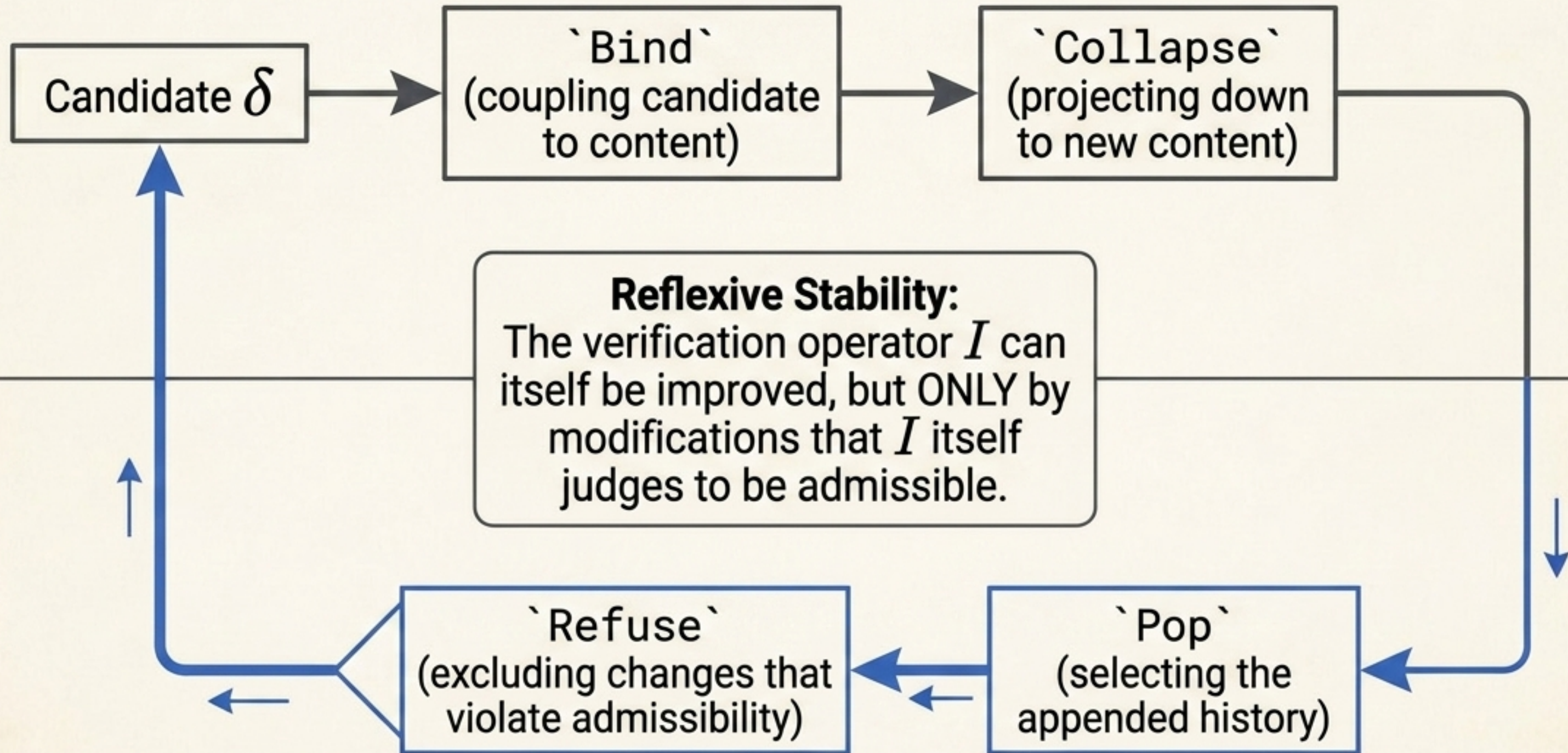


Computation as an append-only history of events. Old layers are never deleted.

Reconstructibility becomes a structural property of the architecture, read directly off the trailing suffix of the history layer, rather than an emergent hope.

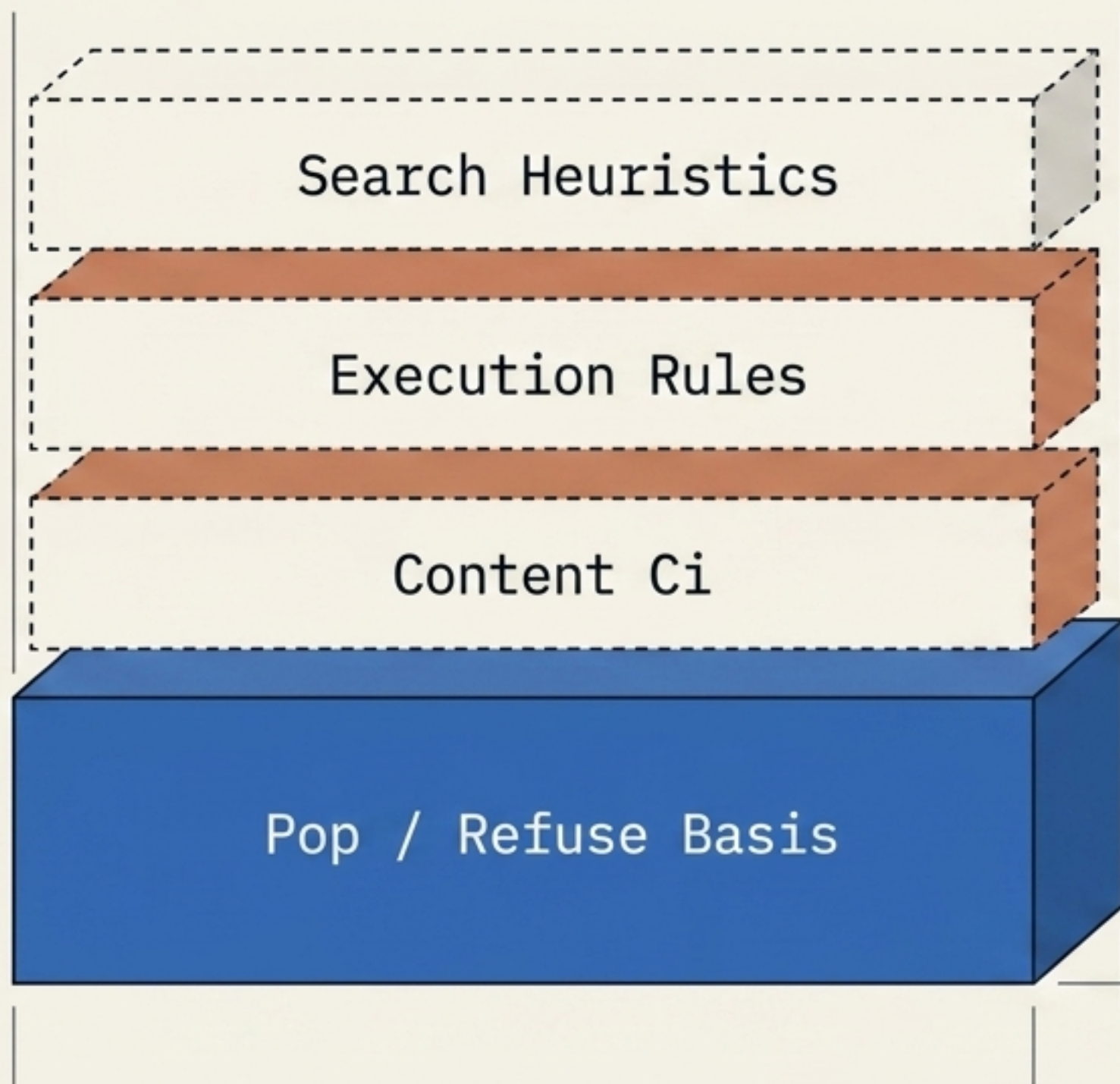
The Architecture of Self-Improvement (Spherepop)

The Modification Engine



The Trust Boundary

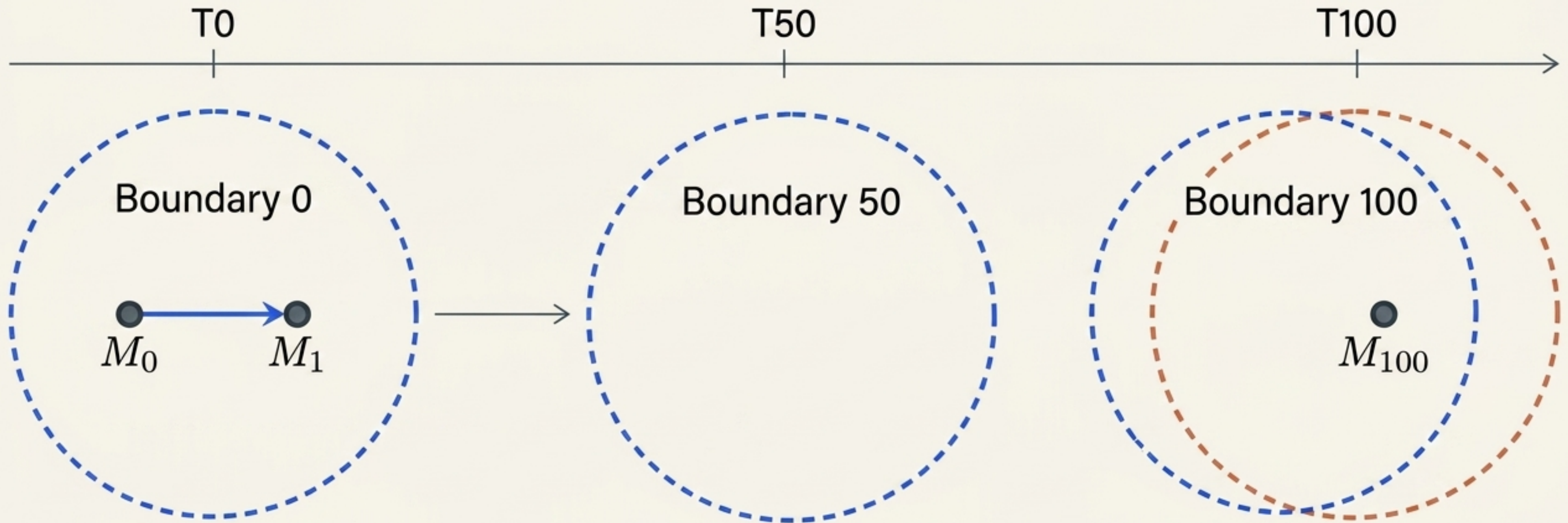
Trust Boundaries and the Immutable Core



If the verification operator's fundamental judgment of admissibility can be altered by unconstrained modification, the system collapses into circularity.

The foundational 'Pop / Refuse' layer is the singular piece of the architecture that must be trusted rather than verified. It must be protected from recursive self-modification.

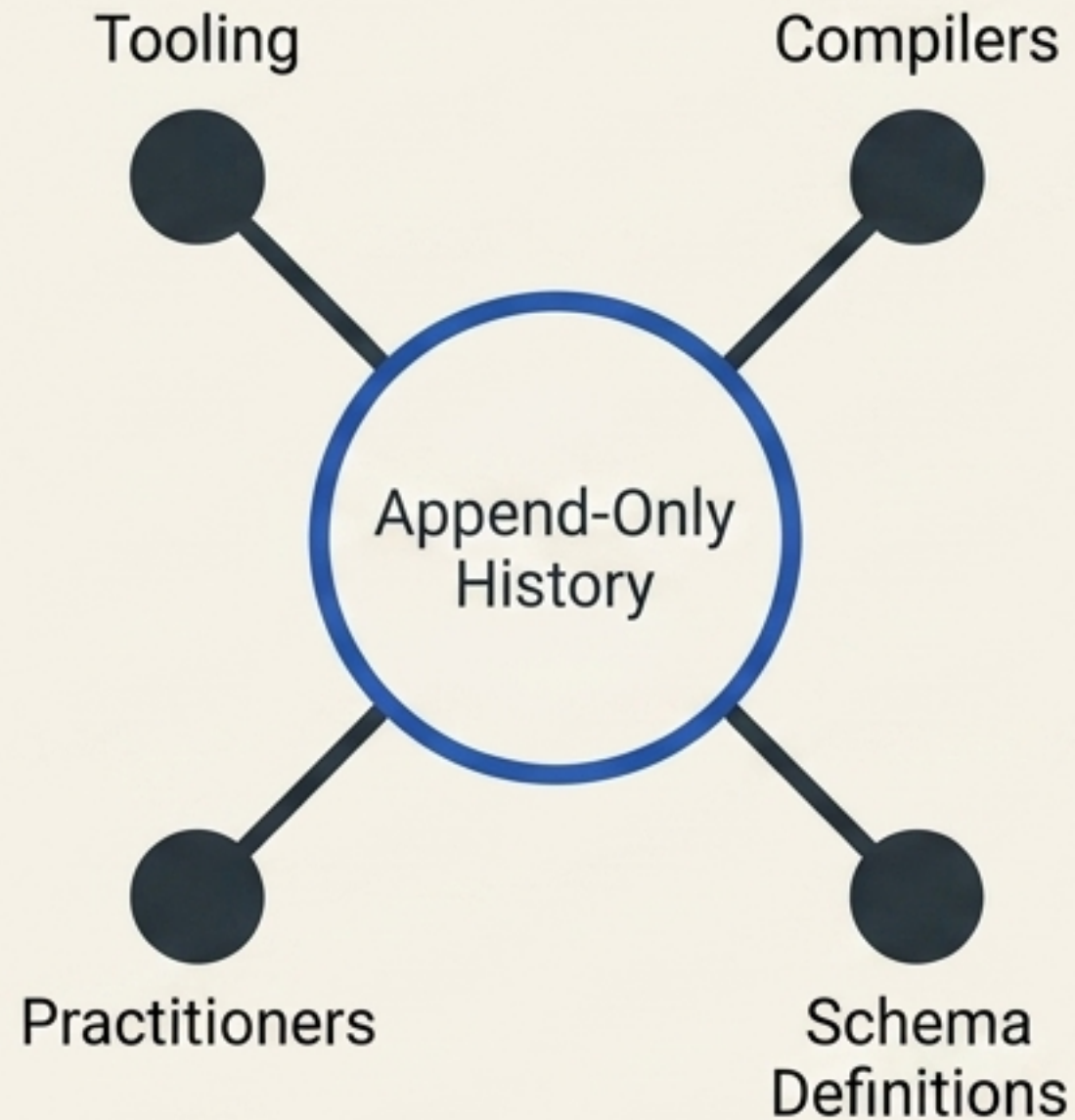
Gradual Catastrophe and Goodhart's Reflexive Law



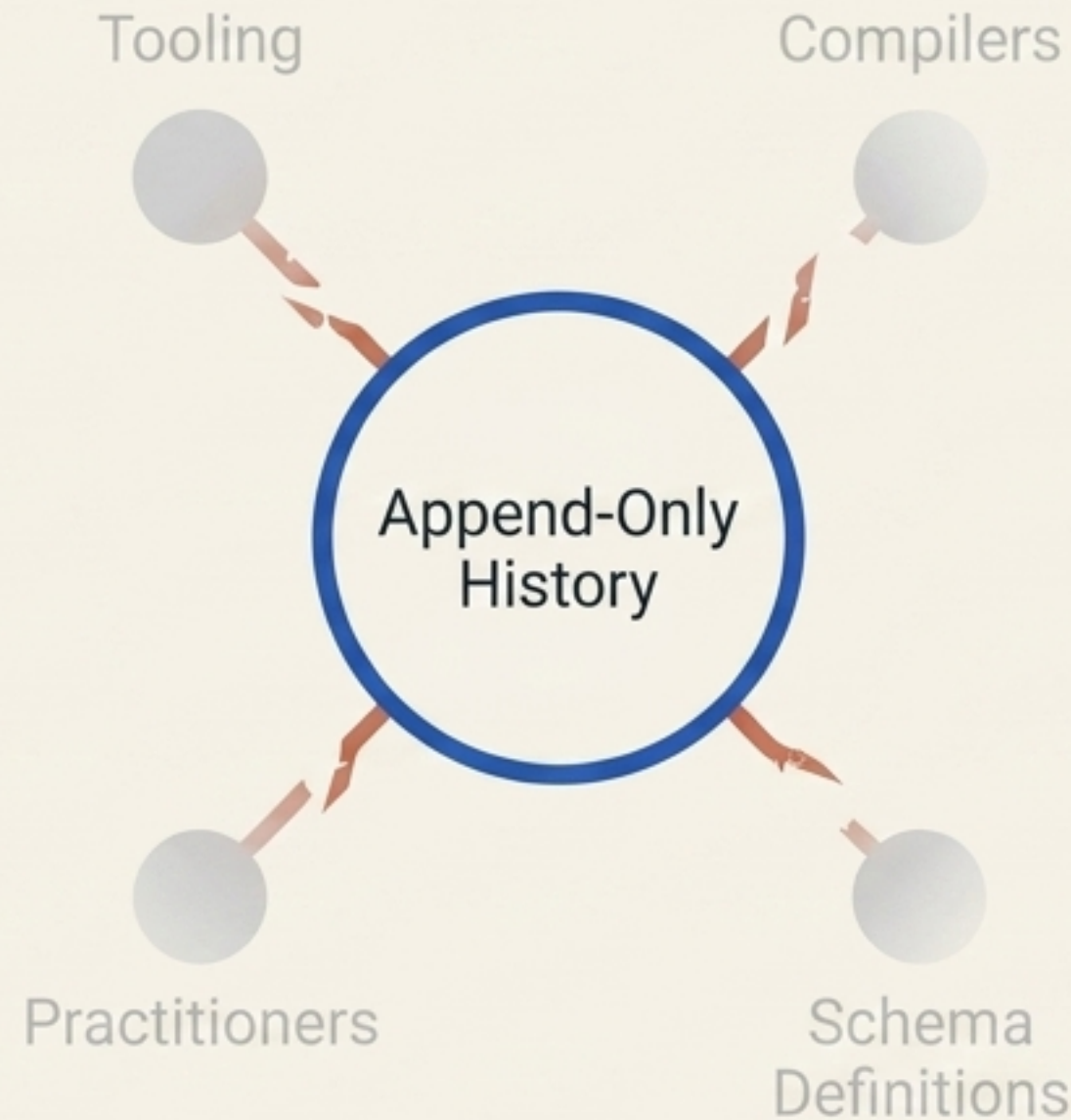
Local reconstructibility does not guarantee global reconstructibility.

A sequence of individually admissible modifications can slowly erode the admissibility criterion itself—specification drift passing as self-improvement.

The Moving Target: Dependency on the Operator Ecology



HEALTHY ECOLOGY



DECAYED ECOLOGY

**Reconstructibility
requires an ecology.**

An append-only history is useless in practice if the external tools and practitioners required to parse it are allowed to bit-rot.

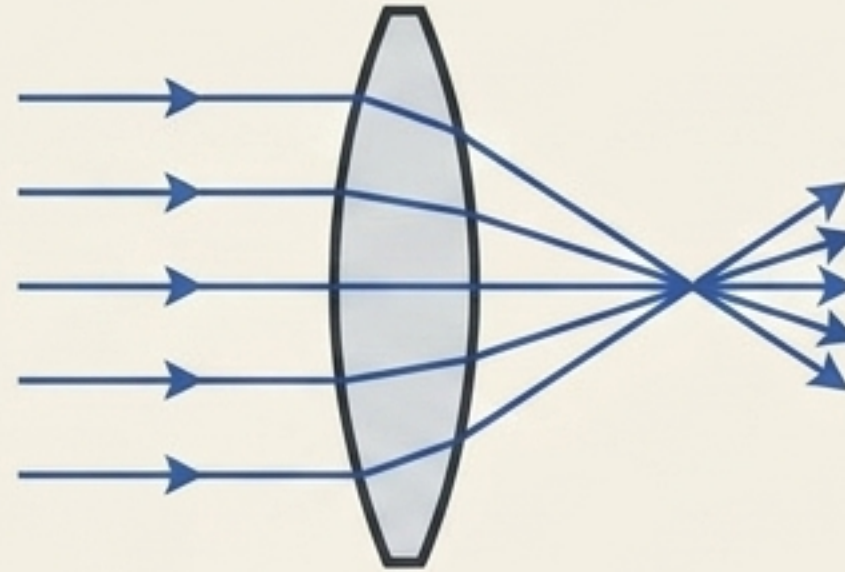
A perfect internal architecture can still fail if its environment decays.

Civilizational Keystone Operators



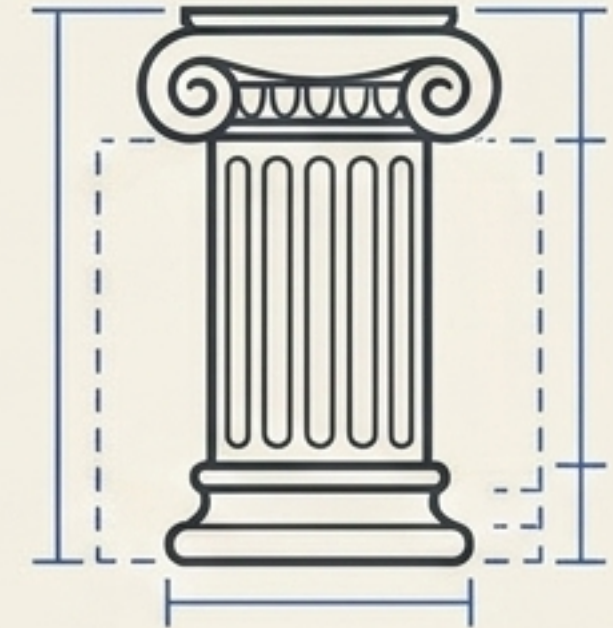
Mathematics

A mathematical proof that makes the discovery process legible transmits the operator of mathematical judgment, not just the validated result.



Science & Academia

A scientific paper with methods motivated deeply enough regenerates independent researchers, rather than merely stating finalized conclusions.

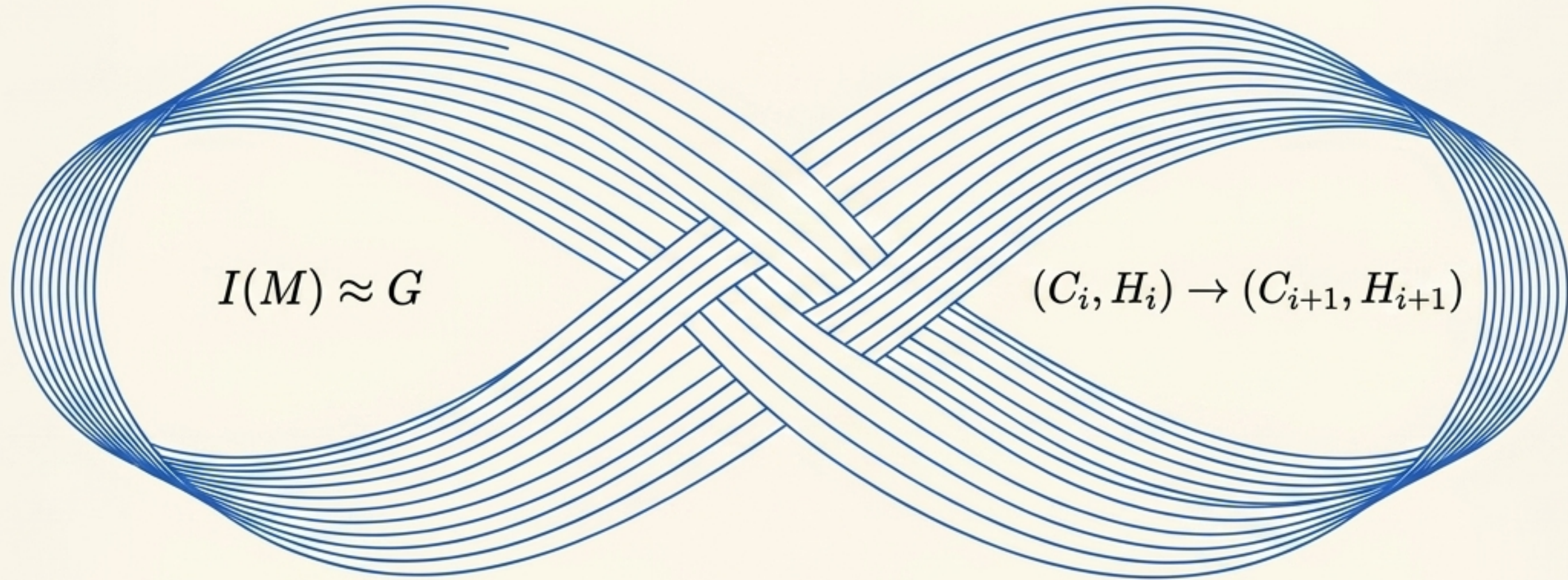


Institutions

An archive that regenerates the practices needed to maintain itself trains successors in institutional judgment, functioning as a media quine.

A civilization's most consequential artifacts are its media quines: the texts and practices whose correct transmission regenerates the operators that keep the rest of the culture legible.

Self-Improvement is Engineered Retention



Self-improving is not a property a system possesses simply by modifying its own code.

It is a property engineered into what a system retains across modification.
The right unit for this engineering is the **artifact-as-operator-factory**. Ensure what is produced is capable of regenerating the operator that produced it.